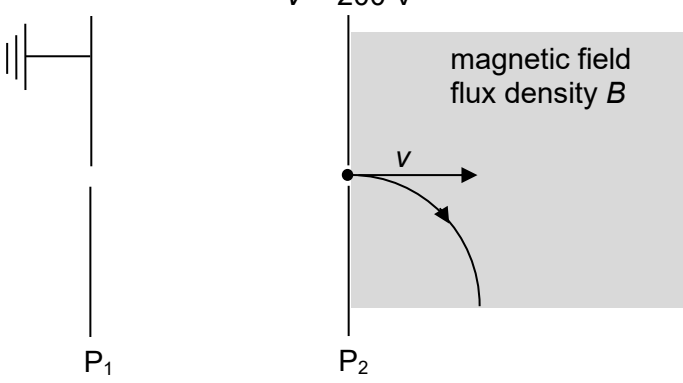


		answer
5(a)	Gain in kinetic energy = Loss in electric potential energy $\frac{1}{2}mv^2 - 0 = eV - 0$ $\frac{1}{2}(9.11 \times 10^{-31})v^2 = (1.60 \times 10^{-19})(200)$ $v = 8.382 \times 10^6$ $\approx 8.38 \times 10^6 \text{ m s}^{-1}$	[1] statement [1] correct answer before rounding off

No.	Solution	Remarks
5(b)(i)	$V = 200 \text{ V}$  The diagram shows two vertical parallel plates, P₁ on the left and P₂ on the right. A battery is connected across them, with the positive terminal at P₁. An electron (represented by a dot) is shown at P₂, moving horizontally to the right with velocity v. A shaded rectangular region to the right of P₂ is labeled 'magnetic field flux density B'. Inside this region, the electron's path is shown as a semi-circular arc curving downwards.	[1] path
5(b)(ii)	For circular motion, $Bqv = \frac{mv^2}{r}$ $B = \frac{mv}{qr}$ $B = \frac{9.11 \times 10^{-31} (8.38 \times 10^6)}{1.60 \times 10^{-19} (2.8 \times 10^{-2})}$ $= 1.7 \times 10^{-3} \text{ T}$	[1] for correct substitution [1] for correct answer
5(c)(i)	The component of the velocity parallel to the magnetic field, is <u>unaffected by the magnetic field</u>. The component of the velocity perpendicular to the magnetic field results in a <u>magnetic force acting on the electron perpendicular to its motion</u>. Combining these 2 effects produces a <u>helical path</u> of the electron.	[1] for correct effect [1] for correct effect [1] correct path
5(c)(ii)	Radius of curvature, $r = \frac{mv_{\perp}}{Bq}$ $= \frac{9.11 \times 10^{-31} (8.38 \times 10^6 \sin 30^{\circ})}{1.60 \times 10^{-19} (1.7 \times 10^{-3})}$ $= 1.4 \times 10^{-2} \text{ m}$	[1] for correct substitution [1] for correct answer

No.	Solution	Remarks
6(a)(i)	If all the electron's energy is absorbed, the electron could reach	